

Clustering Network Traffic by Attack Type

*Final Assignment, Unsupervised Machine Learning course, IBM Machine Learning Professional Certificate
Dataset: NSL-KDD (public network intrusion benchmark)*

Main Objective

This project applies unsupervised learning to network traffic data. The goal is to check if traffic naturally groups into clusters that line up with known attack types, without using any labels during training.

This matters for a security team because most raw traffic has no label attached to it in real time. A model that groups similar traffic together can help an analyst spot suspicious patterns faster, before anyone confirms what the attack actually is.

The approach here is clustering, not dimensionality reduction. PCA is used only as a preprocessing step, to make clustering faster and more stable on high-dimensional data.

Dataset Description

The dataset is NSL-KDD, a public benchmark built for network intrusion detection research. It contains network connection records described by 41 features: protocol type, service, connection duration, byte counts, and various traffic statistics. Each record also carries a label showing whether the connection was normal or which type of attack it was.

For this analysis, the original 22 attack labels were grouped into four standard categories: DoS, Probe, R2L, and U2R, plus normal traffic. The full training set has 125,973 records, but normal and DoS traffic heavily dominate it, and R2L and U2R attacks are rare.

To keep rare attack types visible during clustering, a sample of 4,047 records was built: up to 1,000 records each for normal, DoS, and Probe, and all available records for R2L (995) and U2R (52). This is a deliberate rebalancing choice, not the real-world traffic mix, and it is revisited in the Limitations section.

Category	Records in sample	Share of sample
normal	1,000	24.7%
DoS	1,000	24.7%
Probe	1,000	24.7%
R2L	995	24.6%
U2R	52	1.3%

Data Exploration and Preparation

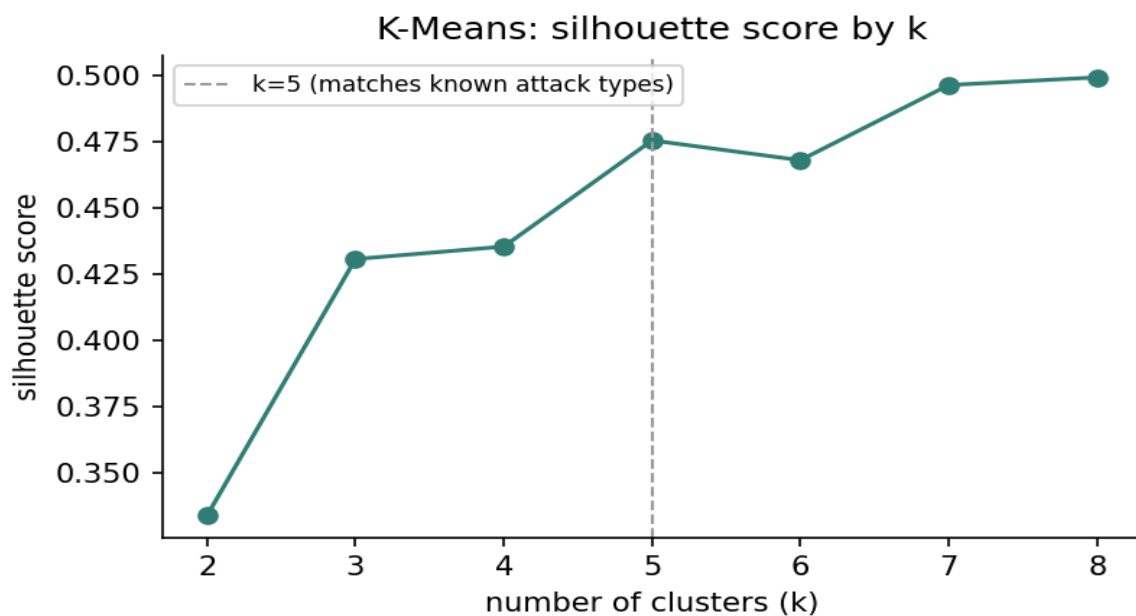
The three categorical features (protocol_type, service, flag) were one-hot encoded. The service feature alone has 70 possible values, so encoding expanded the feature space from 41 to 111 columns. Numeric features were scaled with a standard scaler, so no single feature dominates distance calculations just because of its scale.

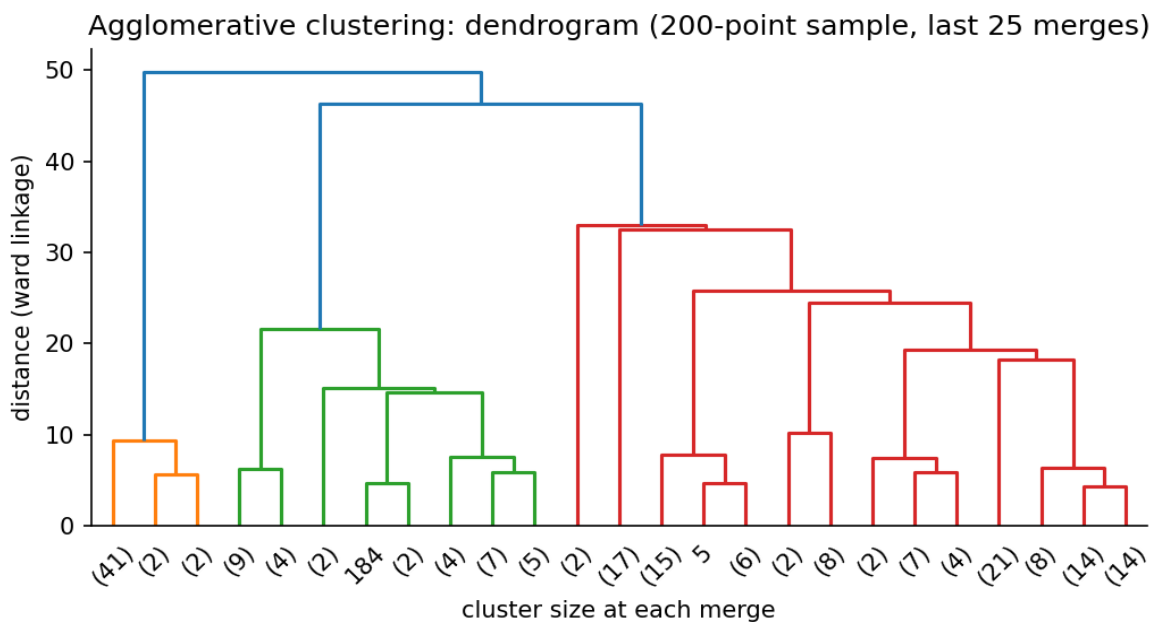
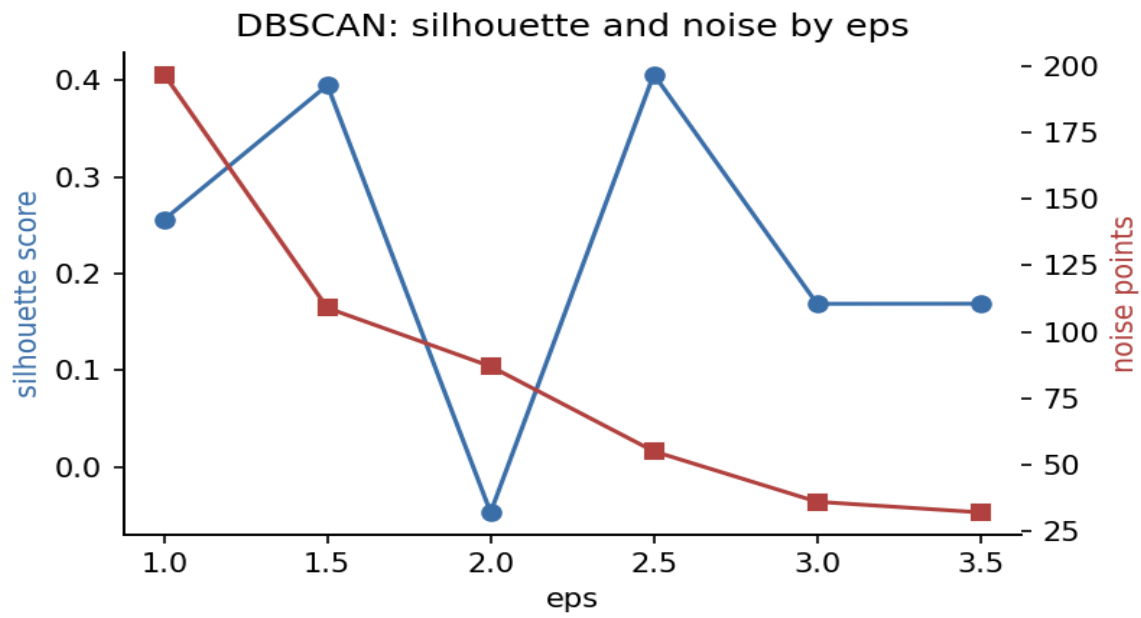
PCA was then applied to the processed data. 17 components were needed to keep 90% of the variance, which confirms the data does not collapse into just a handful of dimensions. A separate 2D PCA projection was also computed for visualization, but it only captures 34.2% of the variance, so the 2D plots below show real structure, not the full picture.

Model Variations

Three clustering approaches were compared, all run on the same 17-component PCA representation, so the comparison is fair.

- **K-Means**: tested with k from 2 to 8, chosen by silhouette score. The score improved steadily up to $k=8$ (silhouette 0.499), without a clear elbow. $k=5$ was also tested since it matches the five known categories, but scored lower (0.476).
- **DBSCAN**: tested with eps from 1.0 to 3.5 ($\text{min_samples}=5$). The best result came at $\text{eps}=2.5$, with 5 clusters, 55 noise points (1.4% of the sample), and a silhouette score of 0.406.
- **Agglomerative clustering** (Ward linkage): run with $k=8$, to match K-Means for a fair comparison. Silhouette score: 0.512, the highest of the three.



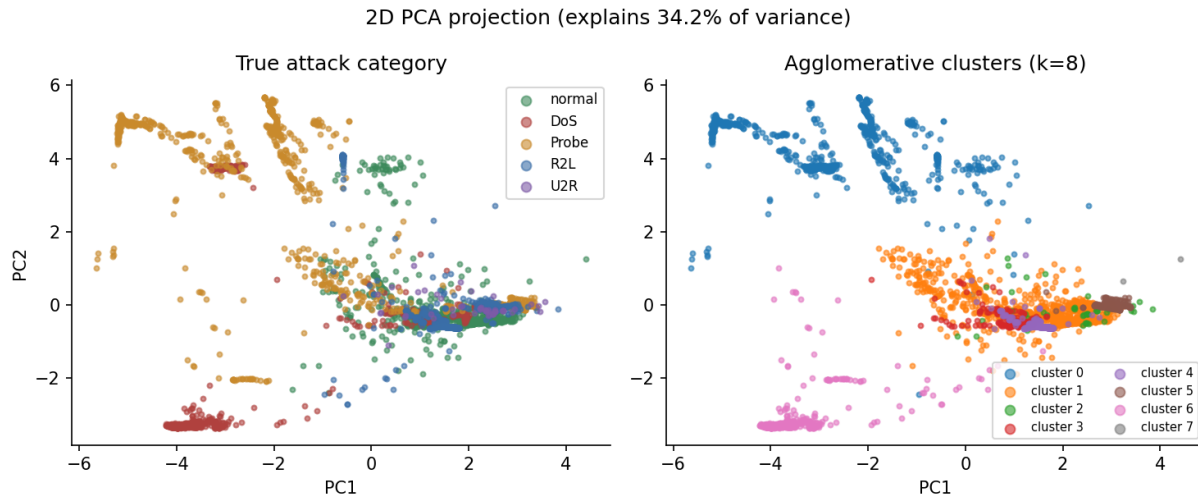


Final Model Recommendation

Agglomerative clustering is the recommended model. It scored highest on silhouette (0.512, versus 0.499 for K-Means and 0.406 for DBSCAN), and it also produced several clusters that line up closely with real attack categories, even though the model never saw a label:

- one cluster was 97.8% R2L traffic
- one cluster was 100% Probe traffic
- one cluster was 93.3% DoS traffic
- one cluster was 100% normal traffic

K-Means gave similar results but scored slightly lower on silhouette. DBSCAN was useful for isolating small pockets of rare attacks (it found a cluster that was 86.4% U2R, the hardest category to detect in this dataset), but it struggled to cleanly separate the larger normal, DoS, and Probe traffic, which sit closer together in density.



Key Findings and Insights

- Several clusters recovered a single attack category almost perfectly, even though the model never saw the labels. This suggests network traffic carries a real, learnable structure tied to attack type.
- Not all traffic separates cleanly. One cluster mixed R2L, normal, and Probe traffic together, meaning some traffic types genuinely overlap in behavior and won't be caught by clustering alone.
- DBSCAN was the only method that isolated a near-pure U2R cluster, the rarest and hardest attack category in this dataset. That is a real practical advantage for hunting rare attacks, even though its overall silhouette score was lower.
- The 2D visualization is honest but limited. Probe and DoS traffic form visibly distinct streaks in the plot, while normal, R2L, and U2R overlap heavily in the middle, because two dimensions can't represent the 17 that the actual clustering used.

Limitations

- The clustering only used 4,047 of the 125,973 available records, capped mainly by the memory cost of agglomerative clustering. Results might shift somewhat on the full dataset.
- The sample was intentionally rebalanced to keep rare attack types visible. This is not the real-world proportion of traffic, so the silhouette scores here should not be read as "real-world accuracy."
- DBSCAN's eps was chosen from six manually tested values, not a formal elbow method on the k-distance graph. A more careful search might change the result.
- NSL-KDD is a well-known benchmark, but it is old and does not reflect current attack traffic. A model that performs well here is not guaranteed to perform well on modern network data.

Next Steps

- Repeat the clustering on the full training set using a method that scales better, such as MiniBatchKMeans, to check if the $k=8$ recommendation still holds.
- Test the chosen model on KDDTest+, the official held-out test set, which includes attack types never seen in training. That is a better check of whether this approach can catch genuinely new attacks.
- Refine DBSCAN's eps using the k-distance elbow method instead of manual testing.
- As a longer-term idea, feed cluster assignments into a SIEM-style detection rule, so traffic landing in a cluster historically tied to attacks gets flagged for review automatically, as an early triage layer before a human looks at it.